

Qatar-1b

R-2316

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120416 · created 2026-08-18T00:34:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456033.87957 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.045
Transit depth (Rp/Rs)^2	3.43 +/- 1.65 [%]
Orbital Inclination (inc)	79.09 +/- 3.2
Airmass coefficient 1 (a1)	1.187 +/- 0.015
Airmass coefficient 2 (a2)	-0.1043 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	3.32
Optimal Annulus	11.47
Transit Duration (day)	0.025 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.045 vs 0.1463 (deviation 0.86 σ)
Duration fit vs published	0.025 d vs 0.0692 d (deviation 1.58 σ)
Mid-time O-C	-0.1 min
Depth SNR	4.1
Residual scatter	1.22 %

Light curve

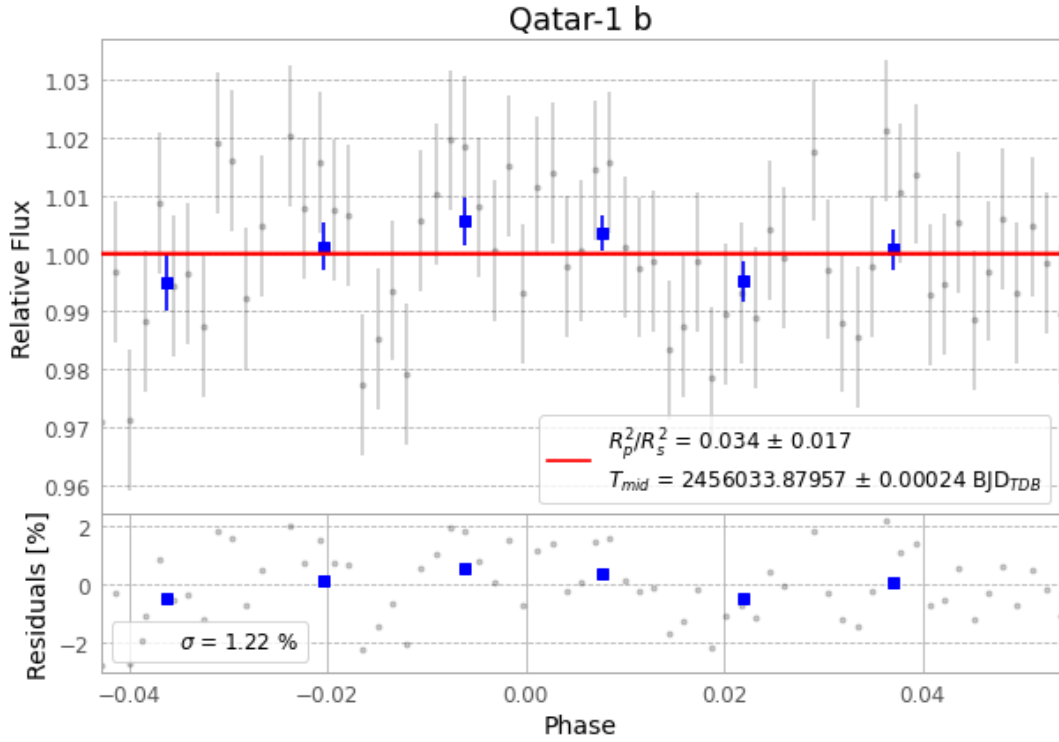


Figure 1 — FinalLightCurve_Qatar-1 b_2012-04-16.png (SHA-256 c9586e30a2c60f39...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 136], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a800f13b115c0cca...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), Qatar-1120416073912.FITS → Qatar-1120416105712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120416.log (b4e563bcb39c...), FinalLightCurve_Qatar-1 b_2012-04-16.png (c9586e30a2c6...), FinalLightCurve_Qatar-1 b_2012-04-16.csv (07688a6c23fa...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:34Z.